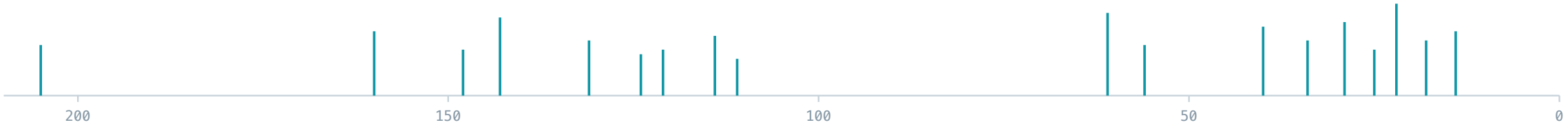
COMPUTER-ASSISTED STRUCTURE ELUCIDATION · FROM NMR

# AILSA

AI + LSD + Agents

An AI agent that aims to autonomously determine the structure of an unknown organic compound from its NMR spectra — reasoning about the data, building solver constraints, and ranking candidate structures without human intervention.




**Domain skill-set**
extensive NMR / CASE expertise & workflow strategy — the intelligence layer

+

**Thin tools**
nmrglue · LSD · RDKit CLI wrappers for data & solvers

+

**Agent team**
**Claude Code's agent-team feature** — specialists that coordinate with each other

= one autonomous CASE system, orchestrated entirely inside **Claude Code**.

**github.com/steinbeck/ailsa**

# Spectra in, structure out — reasoned, not guessed

A chemist elucidating a natural product juggles $^{1}H$, $^{13}C$, HSQC, HMBC and COSY, forms hypotheses, and rejects the ones the data contradicts. AILSA puts that reasoning loop under an AI agent.

The Python tools stay deliberately thin — wrappers around **nmrglue**, **LSD** and **RDKit**. Domain expertise and strategy live in the agent skill, not in hard-coded heuristics.

The hard part

### Under-determined by design

NMR gives connectivity fragments, not a structure. Many candidates fit the same data — the agent must generate, rank, and rule out.

The guardrail

### A team, not a lone agent

A multi-agent architecture prevents unproductive loops and the "clean-but-wrong" answer: low error, plausible, yet the wrong isomer.

The solver

### Emergent, not forced

Rings and skeletons arise from MULT + HMBC + COSY constraints — not from a hand-placed answer. Verified: a benzene ring emerged with zero ring bonds asserted.

THE PIPELINE BRUKER DATA → ANSWER

# From raw Bruker spectra to a ranked structure

01 · READ

### NMR data

1D $^{1}H$ / $^{13}C$ and 2D HSQC · HMBC · COSY read straight from Bruker files.

`nmrglue`

02 · PICK

### Peak picking

Raw picking with SNR floors so weak quaternary carbons survive; the chemist agent applies DEPT/HMBC guidance.

`ailsa pick`

03 · DETECT

### Statistical detection

Hybridisation, forbidden/mandatory neighbours & hetero-hetero bonds from 7.9M HOSE stats.

`ailsa detect`

04 · BUILD

### Constraints + fragments

MULT / HSQC / HMBC / BOND plus DEFF fragment goodlists, written into the LSD file by the LSD-engineer agent.

`lsd-engineer agent`

05 · SOLVE

### LSD structure generator

Nuzillard's deterministic solver enumerates every structure consistent with the constraints.

`LSD (Reims)`

06 · RANK

### $^{13}C$ prediction

HOSE-code shift prediction scores candidates; match-count first, MAE second — no hallucinated fits.

`ailsa lsd rank`

07 · CHECK

### Identity gate

Derive identity from the SMILES via **ailsa identify** — never a recalled name; RDKit-verified.

`ailsa identify`

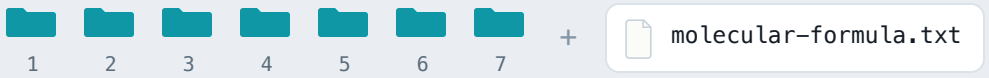
DEREPLICATE

### Known compound?

Optional formula-indexed lookup against 928K compounds runs as a separate path — never inside CASE.

`ailsa dereplicate`

**Input** — one data directory per unknown compound




Numbered **Bruker experiment folders** ($^{1}H$ · $^{13}C$ · HSQC · HMBC · COSY) · plus the known molecular formula

# Four specialists, one coordinator

**case.md · coordinator**

Starts the team, writes the single source-of-truth **CASE-PROGRESS.md**, detects unproductive loops, and intervenes with targeted advisories. Escalates to **ailsa-diagnostic** for deep LSD failure analysis.

spectra

**ailsa-nmr-chemist**

Peak picking, statistical detection, spectral-quality assessment, and the chemistry-first evidence hierarchy — NMR always overrides statistics.

solver

**ailsa-lsd-engineer**

Builds LSD constraint files, iterates HMBC incrementally, owns the constraint inventory (read previous, never reconstruct), runs the solver.

ranking

**ailsa-solution-analyst**

Ranks solutions by $^{13}C$ prediction, judges chemical plausibility, scores confidence, and writes the final result with a tentative identity.

quality gate

**ailsa-devils-advocate**

Validates every LSD file before it runs, checks constraint persistence across iterations, and enforces the G-IDENT / G-MULT gates against wrong-but-clean answers.

**ailsa-lsd-engineer ⇄ ailsa-devils-advocate** · direct review loop — every LSD file is reviewed before the solver runs

**ailsa-lsd-engineer → ailsa-nmr-chemist** · requests — ask the NMR chemist for more correlations or detection results

**Push-never-pull coordination.** The coordinator sends each specialist a `[BEGIN]` after creating its task; agents never poll. From there the specialists that coordinate with each other work through the direct review loop and ask the NMR chemist when they need more data; the coordinator stays the **sole writer** of the progress log.

DATA SOURCE NMRXIV.ORG

# Open, FAIR spectra — fetched by DOI

nmrXiv is an open, community repository for raw and processed NMR data. Every dataset used here — the worked examples and all 256 benchmark datasets — is public, citable, and pulled straight from it — no proprietary files, no vendor lock-in.

## FETCH A DATASET

```
$ ailsa fetch nmrxiv 10.57992/NMRXIV.P9.S51
→ downloads the Bruker experiment folders into ./
```

## HOW IT IS ORGANISED

**Project** *Pn* → **Study** *Sn* → **Dataset** · Bruker

**Why it matters —** openly citable benchmarks let anyone re-run a CASE evaluation on the exact same experimental data.

## WHERE THE CASE EXAMPLES COME FROM

| | | |
|---|---|---|
| P9 | **Teaching Datasets 1** | Duisburg-Essen · CASE1, CASE9 |
| P42 | **Classics in Spectroscopy** | CASE2, CASE4, CASE5, CASE8 |
| P124 | **Alkaloids of Virgilia Oroboides** | Krytovych / Breuning · CASE7 |

CASE3 & CASE6 come from further public nmrXiv datasets (pulegone, citronellol).

BLIND BENCHMARK 256 PUBLIC DATASETS NMRXIV

# Blind on 256 public datasets: 69.1 % solved at rank 1

Measured in a restricted configuration: without the agent team (unattended runs form none, so the specialists ran one at a time under the coordinator) and without statistical detection (outdated reference database on the compute host).

HOW IT IS MEASURED

01

### Spectra in, nothing else

The coordinator gets a sanitised directory of Bruker spectra and the molecular formula, never a name or structure; dereplication is forbidden.

02

### Answers locked away

Every file mapping a dataset to its answer is moved out of reach for the duration of each batch; a system-prompt fence forbids looking for it.

03

### Graded from outside

An independent grader re-derives every candidate's InChIKey with RDKit; the run's own claim is never used.

04

### Constitution only

Stereochemistry is not tested.

**177 / 256**
correct structure at rank 1
69.1 %

**189 / 256**
within the top 10
73.8 %

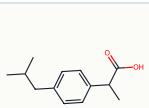
| | | |
|---|---|---|
| ≤ 15 heavy atoms | 40 / 40 | 100 % |
| 16–20 | 28 / 32 | 88 % |
| 21–25 | 49 / 70 | 70 % |
| ≥ 26 | 60 / 114 | 53 % |

Same 102 datasets, two model generations: Opus 5 67 / 102 vs Opus 4.8 40 / 102 at rank 1 — won 30, lost 3, exact McNemar $p = 1.4 \cdot 10^{-6}$

**The limit is size.** Up to 15 heavy atoms every molecule was solved, from 26 upward about half. When a run fails, the right structure is almost never further down its list — generation, not

# Nine worked examples

Illustrative public datasets, separate from the benchmark. Ground truth derived independently (nmrXiv + $^{13}C$ verification); each run graded by a fresh blind instance, RDKit-verified.

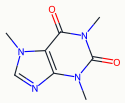


CASE1 Solved

## Ibuprofen

$C_{13}H_{18}O_2$ nmrxiv P9 / S51

Benzene ring emerged from constraints — 0 ring-BONDs; Rank 1, exact InChIKey

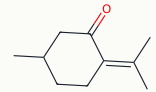


CASE2 Solved

## Caffeine

$C_8H_{10}N_4O_2$ nmrxiv P42 / S1328

Purine alkaloid; two C=O + three N-$CH_3$ pin the xanthine core

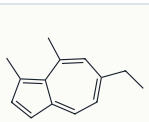


CASE3 Solved

## (R)-Pulegone

$C_{10}H_{16}O$ nmrxiv pulegone set

α,β-unsat. ketone; $^{13}C$ rules out the formula-twin thujone

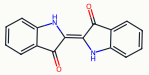


CASE4 Partial

## Chamazulene

$C_{14}H_{16}$ nmrxiv P42 / S1312

Correct azulene scaffold — exact substitution regiochemistry not yet reachable

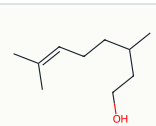


CASE5 Solved

## Indigo

$C_{16}H_{10}N_2O_2$ nmrxiv P42 / S1302

$C_2$-symmetric; single C=O rules out asymmetric indirubin

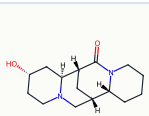


CASE6 Solved

## (R)-Citronellol

$C_{10}H_{20}O$ nmrxiv citronellol set

Acyclic monoterpenoid; olefinic pair + $CH_2OH$ rule out menthol

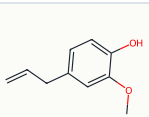


CASE7 Solved

## Virgiline

$C_{15}H_{24}N_2O_2$ nmrxiv P124 / S1266

Tetracyclic lupin alkaloid; independently re-validated on a clean host

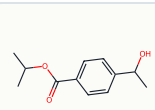


CASE8 Solved

## Eugenol

$C_{10}H_{12}O_2$ nmrxiv P42

Allyl-phenol; terminal =$CH_2$ rules out isoeugenol



CASE9 Solved

## Isopropyl 4-(1-hydroxyethyl)benzoate

$C_{12}H_{16}O_3$ nmrxiv P9 / S53 (D202)

First-ever full solve (2026-06-11); benzoate ester, MAE 1.17

BY THE NUMBERS SCALE & PROVENANCE

# What the agent stands on

**928K**
reference compounds
COCONUT + NMRShiftDB

**7.9M**
HOSE statistics
$^{13}$C shift prediction

**2.4M**
fragment SSCs
DEFF goodlist search

**111K**
unique formulas
formula-indexed lookup

**1372**
passing tests
pytest · mypy-strict · ruff

**69.1%**
rank 1 in the blind benchmark
177 / 256 · restricted configuration, see slide 6

**16**
CLI command groups
every one --format json

**5**
agents in the team
coordinator + 4 specialists

## BUILT ON

**LSD** — the deterministic structure generator by Jean-Marc Nuzillard (Université de Reims).
**HOSE codes** — Bremser, 1978. Parsing via **nmrglue**; cheminformatics via **RDKit**.

## GET IT

**github.com/steinbeck/ailsa** · **pip install ailsa** · reference database DOI **10.6084/m9.figshare.31073554**